



SONAM SONI
Engineering Lead

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Area of Interest

- Artificial Intelligence, Deep Learning, NLP, Generative AI, LLM,
- LSTM, RNN, GPT, Open AI, Python

Interests

- Painting
- Cooking
- Dancing
- Movies & Music

Education

- Aug 2018 – 2024 **Doctor of Philosophy, Computer Science (NLP, AI)**
Amity University, Noida
Thesis submitted
- Jul 2014 – May 2016 **Masters of Technology, Computer Science**
MPSTME, Mumbai
CGPA: 2.9/4
- Jul 2010 – May 2014 **Bachelor of Technology, Computer Science**
United Institute of Technology, Allahabad
Percentage: 68%

Work Experience

July 2025 – Present **Lead (Artificial Intelligence)**
Kingsley Gate

Project: AI-Powered Candidate Search Platform – Quick & Extended Search

- Architected and implemented a real-time candidate recommendation system delivering ranked, explainable candidate results within 60 seconds using semantic expansion and multi-source querying.
- Designed LLM powered title & company expansion, improving search coverage and recall by generating synonyms, role variants, and related company mappings.
- Built a multi-stage scoring & filtering pipeline using LLM-based explainability, enabling transparent and recruiter-trustworthy recommendations.
- Skills: Artificial Intelligence, ML, MLOps, Python, pytorch, TensorFlow, Hugging Face, OpenAI GPT, Gemini, LLaMA, NLP, Docker, Postman, FastAPI, AWS sagemaker, S3, GenAI, BERT, Transformer, GPT.

Feb 2024- July 2025 **Engineering Lead (Artificial Intelligence)**
Persistent System

Project: Knowledge Search (United Health Group)

- Working on developing a scalable text classification system capable of accurately categorizing healthcare texts, improving organizational efficiency and decision-making processes using the concept of Large language model LLM, GPT, open AI, BERT.
- This project showcases my proficiency in Python programming, machine learning, and data handling techniques in the context of healthcare information management.

Project: Semantic Modelling (United Health Group)

- Working on implementing a state-of-the-art semantic modeling framework using the concept behind GPT models, BERT, LLM, Transformer, enhancing the ability to extract, understand, and utilize healthcare information.
- This project underscores my expertise in advanced NLP techniques, particularly in semantic modeling and healthcare informatics, showcasing my ability to apply cutting-edge technologies to solve complex challenges in the healthcare domain.

Project: NLP Voice kit (Apple)

- Worked on API development, NLP deployment of project of NLP voice kit chatbot project, the project focuses on initiating a search operation for a specific query given by user. Used various customized functions for API development of the query as well as inbuilt libraries of python like NLTK, pandas, numpy, Requests, Flask.
- Used Solr and mongo DB for data set accessing and storing using the pre- processed data using python framework.
- Used various pre-processing approaches for refining the dataset, as the data received from client was in unstructured format. (Includes telephonic conversation and chat of the customer and user)
- The API uses LTR and BM25 ranker algorithm for calculating the score. Also worked on Boosting on the above search operation using various customized approaches for reducing the response time.

Project: Apple Sales Forecast Analysis (Apple)

- Working with apple to predict future customer demand for products or services of apple, used several powerful python libraries and tools that can be used for demand forecasting analysis i.e. NumPy, pandas, and scikit-learn for time series forecasting doing Demand Forecast Analysis of the products using historical data, doing prior analysis of a products like Order Shortage, Low inventory, Continuous decline of sales of a products using Machine learning approach.
- Worked on Data Preparation, Data Visualization understand the Patterns and trends in the demand, Time Series Analysis:
- Performed time series analysis on the data, such as identifying seasonality, trend, and any other relevant patterns.
- Worked on finding an appropriate forecasting model, such as ARIMA (Autoregressive Integrated Moving Average) or SARIMA (Seasonal ARIMA), based on the characteristics of the data. Splited the data into training and testing sets to evaluate the model's performance.
- Trained the selected forecasting model on the training data. Evaluated the model's performance on the testing data using appropriate metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), or Root Mean Squared Error (RMSE). Used the trained model to make predictions for future demand.
- Used Snowflake for data set accessing and Data lab for python script development. Using various python libraries for Exploratory Data Analysis using Dtale, Pandas etc.

Jun 2020 –
March 2021

Machine Learning Engineer
Chefling Inc, Bangalore

Project: Personalized Recommender System

- Worked on building a recipe-based recommender system using hybrid approach incorporating conventional techniques, content as well as collaborative filtering.
- Devised Matrix Factorization, Topic Modelling, Named Entity Recognition with spacy as NLP framework with single optimized functions for each recommendation task implementation. Experience with API Architecture using Flask with deep understanding of the overall development process and software delivery lifecycle.

Project: Ingredient Content Extraction and Elmo embeddings

- Worked on data scrapping to generate dataset for ingredient content extraction. Used BeautifulSoup on all scrapped links of ingredients content which was converted in to objects and contents are accessed by specific HTML tags and classes.
- Embedding from Language models (Elmo) technique is used on extracted data to train deep pre train neural network after data cleaning and wrangling using Keras library.

Ph.D Project:

AI powered Automatic Question and Answer Generation from test for learning Use cases.

- Worked on generating Automatic Question generation like factoid type, gap fill Question, using various Word embedding techniques for natural language processing (NLP) tasks, converted words or phrases into numerical representations that capture semantic relationships and context i.e. BERT, ELMOW, seq2seq, word2vec, GLOVE.
- Used Various techniques like, RNN, LSTM, BI-LSTM, Large Language Model (LLM) for the same.
- Used Various datasets for evaluating the results of the algorithm i.e. Squad, Natural Question, QuAC.

Assistant Professor

Jul 2016 - July 2019: TCECT College, VJTI, Rizvi College, IMS Engineering College

- Provided Teaching Support for graduate level course, "Computer Graphics, Artificial Intelligence, Machine Learning, Python Electronic Circuit & Communication Network, Human Machine Interaction, and Computer Networks".
- Responsible for advising students, curriculum development, and grading papers.

Live Projects on Artificial Intelligence

Project: Churn Prediction in an Organization

Organization: Bennett Technologies, Gurgaon

- Applied the decision tree algorithm on the data set provided from the company and predicted the probability of an employee retention.
- Used Amazon RDS by AWS to retrieve data for applying the algorithm using Python and SQL queries.
- Predicted the churn rate by dividing the data set in training and testing with the accuracy of 82%.

Project: Recommender System in Health Care

Organization: University of Western Sydney, Australia

- Working on collaborative filtering and context-based filtering to recommend health related information to a patient
- The goal of the health care recommender system is to supply its user with medical information which is meant to be highly relevant.

Membership

- Member of International Association of Engineering (IAENG).
- Member of Universal Association of Computer and Electronics Engineers i.e. UACEE.

Research & Publication

- Automatic Question and Answer Generation from text using **Neural Networks** in conference **FICR-TEAS**.
- Automatic question-answer generation from video lecture using neural machine translation in **SPIN**
- Automatic Fill in the blank type of Question-and-Answer Generation from Text using NLP in Journal **'JARDCS'**.
- Automatic Question Generation: A Systematic Review in **'ELSEVIER'** Conference.
- Road Map to Visual Cryptography in **'IJARCSSE'**
- Information Hiding using the concept of Visual Cryptography in **'IJSER'**